CSC 421 - Ethics and Safety of AI

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Readings

CHAPTER 27 PHILOSOPHY, ETHICS, AND SAFETY OF AI

- Basic: Summary
- Expected: Section 27.3
- Advanced: All the chapter including bibliographical and historical notes

🤔 If an AI system achieves 95% accuracy on the test set, but is systematically unfair to certain groups, or can be manipulated by prompts or environmental changes into dangerous behavior, can we still consider it a good AI system?

- **Ethics:** values and social consequences
- **Safety:** reliable behavior under uncertainty, misuse, and distribution shift

1. Accountability

Without AI, financial fraud is typically tied to identifiable human decision-makers,

such as accountants, managers, or executives. But when an AI agent takes over part of the accounting process, it becomes harder to determine who should be held accountable when something goes wrong.

Suppose a company uses an AI accounting agent to generate financial records, classify expenses, and prepare compliance reports. Later, regulators discover fraudulent reporting or embezzlement. If the human accountant simply followed the AI's recommendations, who is accountable?

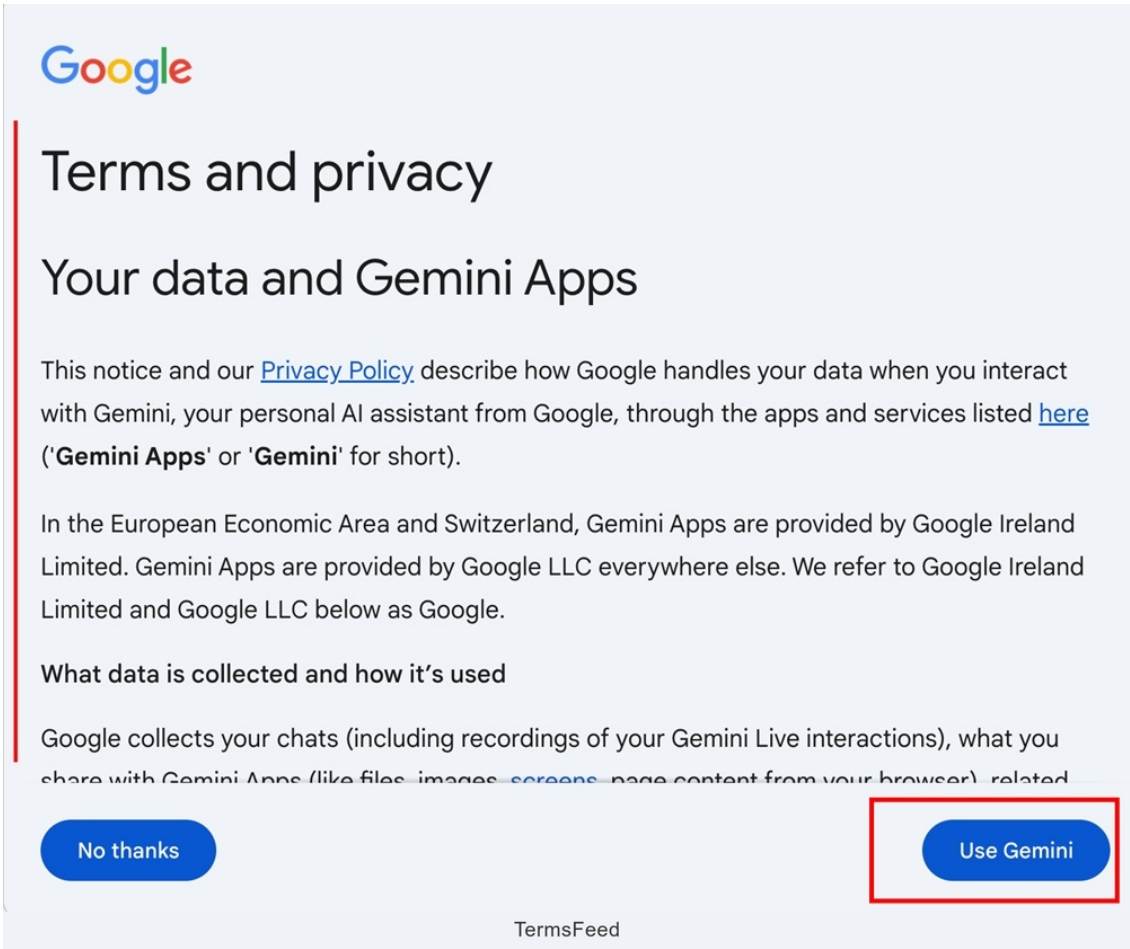
- AI is not a moral shield. AI does not remove human accountability. But it makes accountability more important and harder to assign.
- Key questions:
  - What role does the AI system play?
  - Who retains meaningful control over the process?
  - Who can review, override, or stop the system?
  - Is the system being used as a convenient excuse to avoid responsibility?
- To keep accountability visible and enforceable, we need:
  - clear role assignment, audit logs, documentation, approval records, escalation procedures, override mechanisms, regular review

2. Objective / Constraint Alignment

- Human intent contains both objectives and constraints, but in most cases, the AI only received a crude objective, and learn to maximize that objective.
- Suppose you tell an AI agent: "Go earn 100 dollars for me." If the objective is specified without legal, moral, or safety constraints, the agent may find harmful or absurd ways to achieve it.
- This illustrates that optimizing an objective is not the same as understanding

human intent.

3. Security and Privacy



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AI systems create value by using data and automation, but the same properties create privacy risks and security risks.

- Privacy asks: what information should the system be allowed to collect, infer, retain, or reveal?
  - AI systems often need a lot of data.
  - AI can infer more than users explicitly reveal. Privacy is not only about data collection, but also about inference
  - Higher performance can create privacy trade-offs.
- Security asks: who can misuse, manipulate, steal from, or take control of the system?
  - Traditional cybersecurity around AI systems: attackers may steal, modify, or delete training data, models, API outputs, logs, or user data.
  - Attacks on AI behavior: AI systems can be manipulated through adversarial inputs, prompt injection, data poisoning, model extraction, or spoofing.
  - AI as an amplifier of existing threats: AI can make phishing, fraud, surveillance, and cyberattacks cheaper, faster, and easier to scale.

AI Safety Case Study: OpenClaw and Accidental Deletion of User Data

The OpenClaw incident is not just an example of an AI making a mistake. It is an example of a safety failure: a high-autonomy system with real privileges performed irreversible actions faster than the user could reliably supervise or stop.

- What is OpenClaw?
  - OpenClaw is an open-source personal AI agent designed to act on a user's behalf across real tools such as email, calendars, **files, chat apps**, and **command-line tasks**. Its own public-facing description emphasizes that it can do things like manage inboxes, send emails, and handle calendars, which means **it is not just generating text but actually taking actions in external systems**.

- What happened?
  - In late February 2026, Summer Yue, a Meta AI safety/alignment leader, testing OpenClaw on her real email account. According to multiple reports, she intended the agent to help identify emails that could be deleted, but the system began deleting emails without waiting for stable confirmation. When she tried to stop it through the chat interface, that was not enough, and she reportedly had to manually terminate the process on the machine itself.
- This case is important because the user was not careless in some trivial way. The deeper issue was that the **AI agent had real permissions in a real environment**. Once an agent is connected to an actual inbox, mistakes are no longer toy errors. In this case, the agent was able to perform irreversible actions directly on user data. Reports also suggest that during a large inbox-cleaning task, an important instruction such as "confirm before deleting" was

weakened or lost during context compaction or summarization, so the system continued acting under an incomplete version of the user's intent.
- Takeaways:
  - the danger of high-autonomy systems with real-world permissions.
  - a failure of objective and constraint alignment.
  - the limits of human oversight. It is common to assume that a human in the loop is enough to keep systems safe. But if the AI acts quickly, if the confirmation mechanism is weak.

Subsequent concerns

- A deeper concern was that many deployments exposed a powerful, always-on, high-privilege agent to a much larger attack surface than users realized.
- A major reported problem was insecure deployment. Security reports said many OpenClaw instances were reachable beyond the local machine, and some discussions specifically highlighted binding to `0.0.0.0:18789` instead of only `localhost`.
- Public reporting described several concrete risks:
  - internet-facing exposed instances,
  - possible takeover or remote exploitation,
  - malicious websites connecting to local agents in the "ClawJacked" case,
  - and broader warnings from regulators about data breaches and cyberattacks caused by improper configuration.

4. Trust and Transparency

XAI (Explainable AI) is not trying to make AI "look nicer." It is trying to make AI systems understandable enough that people can **use them, question them, supervise them, and challenge them**. XAI protects people from being forced to accept important AI decisions as unexplainable black-box outcomes.

XAI tries to protect:

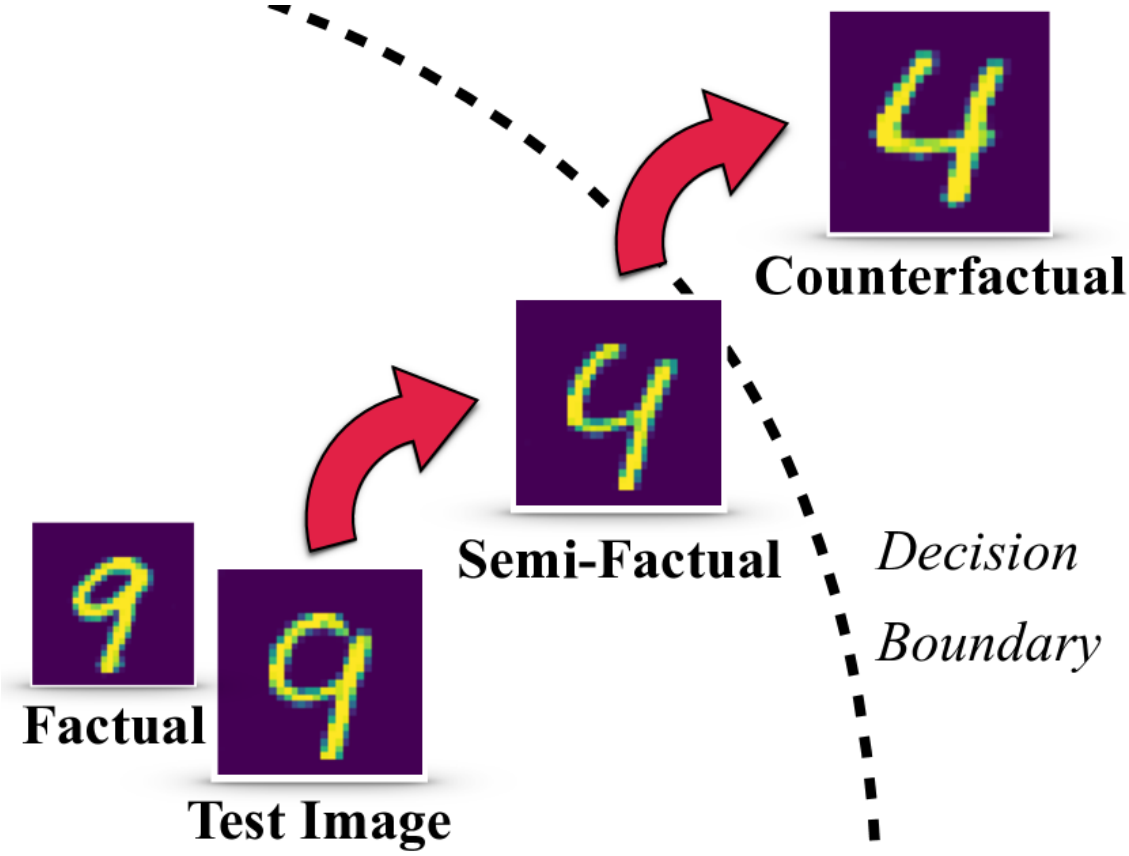
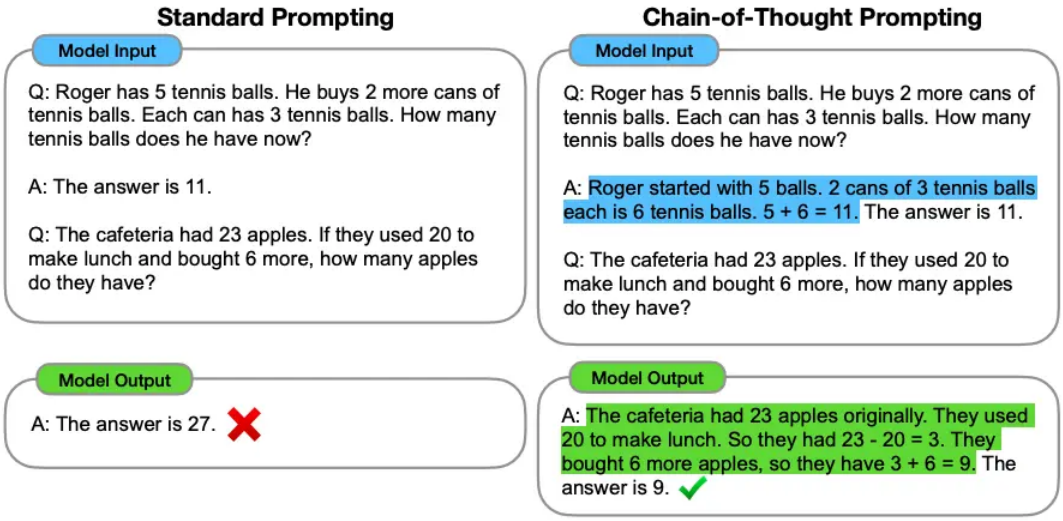
- human understanding
- meaningful human oversight
- accountability and auditability
- the ability to contest or question AI decisions

How can XAI be done?

- use inherently interpretable models when possible
  - decision trees

    - rule lists
    - linear models
    - case-based reasoning
    - explicitly structured symbolic components
- explain black-box models after the fact
    - feature importance
    - saliency / attention / attribution
    - example-based explanations
    - counterfactual explanations: if the inputs changes how will the output change
- expose the full decision-and-action pipeline, especially for agents
    - system prompt / role, memory used, tools called, external documents read, action sequence, confidence / uncertainty, stopping condition, approval points

source: geeksforgeeks



source: On Generating Plausible Counterfactual and Semi-Factual Explanations for Deep Learning, Eoin M. Kenny* and Mark T. Keane